45 Open Source Landscape Analysis and Suitability Modelling in GIS

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Public bodies at both the European and national levels are making increasing amounts of environmental data openly available. Although these datasets were not produced with archaeological research in mind, they offer considerable potential for investigating past landscapes and human land use. This paper reviews the most relevant open-access environmental datasets and demonstrates how they can be integrated into an open-source GIS methodology for landscape analysis and agricultural suitability modelling.

The approach combines freely available soil data with topographic variables, including slope, aspect, and elevation, along with temperature data within a multi-criteria analysis framework to reconstruct past land use and evaluate agricultural potential. As a case study this paper applies this approach to the study of medieval Tuscany, focusing on the region of Lucca to investigate patterns of rural land use. It develops suitability models for wheat, grapevine, and olive cultivation, and evaluates these models against settlement patterns between AD 1000 and 1260. The analysis explores how land-use practices changed across the region over this period, and their contribution to broader socioeconomic developments.

The methodology presented here is fully reproducible, relies on entirely open-source data and is designed to be tested against independent lines of evidence, including paleoenvironmental, archaeological, and historical data. In addition, this workflow is transferable to other geographical regions and chronological periods, and can be adapted according to data availability, spatial scale, and specific research questions, providing a flexible and transparent open-source methodology for landscape archaeology.